

Scheme 1

Stage 1 Screening and Eligibility Decision Scheme

Systematic Review on How the Biopsychosocial Model Frames Chronic Pain Research

Purpose

This scheme operationalizes title and abstract screening after search and deduplication. Its function is to decide whether a record should enter the review corpus for downstream coding, using a human-validatable rule set centered on biopsychosocial language, chronic pain relevance, review design, and population eligibility.

Operational Status

- Workflow position: pre-extraction eligibility screen.
- Operational mode: rule-based provisional machine assist with human validation.
- Canonical implementation basis: executable screening rules plus protocol decision guidance.

Canonical Source Paths

- `src/protocol/decision_rules/screening_rules.md`
- `src/bps_review/screening/rules.py`
- `src/review_stages/03_screening/README.md`
- `src/protocol/osf/OSF_registration-HTBMFCPR.md`

Inputs Reviewed

title	Record title.
abstract	Record abstract text.
publication_types	Publication type metadata used to distinguish reviews from protocols, commentaries, and primary studies.
language	Language metadata used to enforce English-language eligibility.
publication_date / year	Publication timing, checked against the operational search window.

Decision Categories

stage1_decision	Implemented values are include , exclude , and unclear . The prose rules also allow a borderline maybe state; in the current executable implementation, ambiguous cases are operationalized as unclear .
stage1_reason	Controlled reason attached to exclusions or unclear cases.
stage1_confidence	Implemented values are high , medium , and low .
stage1_screened_by	In current outputs this is recorded as codex_machine_assist .
stage1_screening_mode	In current outputs this is recorded as rule_based_provisional .

Inclusion Logic

- The record is a review article or review-like evidence synthesis.
- The title or abstract explicitly contains a biopsychosocial term: **biopsychosocial**, **bio-psycho-social**, or **bio psycho social**.
- The focus concerns chronic pain, persistent pain, or a chronic pain condition.
- The population is adult or mixed-age rather than pediatric-only.
- The record is within the operational search window and in English.

Exclusion Logic and Implemented Reason Labels

no biopsychosocial term in title/abstract	No explicit biopsychosocial term is present in title or abstract.
outside operational search window	Publication date falls before or after the operational search dates configured in config/protocol.yaml .
protocol	Protocol papers are excluded.
commentary/editorial/letter	Commentary-type publications are excluded.
animal/non-human focus	Animal-only or non-human records are excluded unless a human focus is explicit.
pediatric-only focus	Pediatric-only populations are excluded.
acute pain focus	Acute pain records are excluded when chronicity is not also explicit.

chronic pain focus unclear

Pain is present but chronic pain relevance is insufficiently clear.

non-English record

Non-English records are excluded.

review status unclear

Used when the record cannot be confidently identified as a review or evidence synthesis from title, abstract, or publication metadata.

Borderline Handling

- Ambiguous review type, mixed acute/chronic populations, unclear age group, or unclear pain duration should not be over-excluded.
- Borderline cases are logged with a rationale and lower confidence.
- Musculoskeletal ambiguity is intentionally carried forward; Stage 3 full-text adjudication resolves that ambiguity rather than Stage 1 excluding too aggressively.

Primary Outputs Using This Scheme

- `src/review_stages/03_screening/outputs/stage1_screening.csv`
- `src/review_stages/03_screening/audit/stage1_screening_summary.csv`
- `src/review_stages/03_screening/audit/reliability_report.csv`

Interpretive Note

Although this scheme is not prompt-driven, it is part of the operational coding architecture of the review because it creates the standardized eligibility categories that determine which records can enter Stage 2 abstract coding and, later, Stage 3 full-text adjudication.